Keystroke Anomaly Detection with the Mahalanobis Distance

DS5020 Final Project, Iteration 2: The Math Foundation

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1 Overview

We want to tell whether a password was typed by its owner or by someone else, using only the timing of the key presses. For one user, we model the timing vector of each attempt as $x \sim \mathcal{N}(\mu, \Sigma)$ and estimate μ and Σ by maximum likelihood from that user's own attempts. A new attempt is flagged as an anomaly when its squared Mahalanobis distance from $\hat{\mu}$ is larger than the $(1 - \alpha)$ quantile of a χ^2 distribution. The real data are the CMU Keystroke Dynamics Benchmark (Killourhy and Maxion, 2009).

One fact about the data affects the whole method: the 31 timing columns are exactly linearly dependent, so the 31×31 sample covariance has rank 21 and cannot be inverted (Proposition 1). We therefore drop 10 columns and work with $p = 21$ features.

2 Notation

Symbol	Meaning	Dimension
Data		
n	number of training attempts of one user	scalar
m	number of new attempts to score	scalar
p	number of features ($p = 21$ for the real data; $p = 5$ in the simulation)	scalar
H, DD, UD	hold, down-down and up-down timing columns, in seconds	11, 10, 10 columns
x_i	timing vector of the ith training attempt	$\mathbb{R}^p$
X	training data matrix of one user; row i is $x_i^\top$	$\mathbb{R}^{n\times p}$
x	a new attempt	$\mathbb{R}^p$
Y	matrix of new attempts; row j is one new attempt	$\mathbb{R}^{m\times p}$
$\mathbf{1}$	vector of ones (its length follows from the context)	$\mathbb{R}^n$, $\mathbb{R}^m$ or $\mathbb{R}^p$
Model and estimates		
μ, Σ	true mean vector and covariance matrix of the genuine user	$\mathbb{R}^p$, $\mathbb{R}^{p\times p}$
$\bar{x}$	sample mean $\frac{1}{n}\sum_i x_i$	$\mathbb{R}^p$
$\hat{\mu}$, $\hat{\Sigma}$	maximum-likelihood estimates: $\hat{\mu} = \bar{x}$, $\hat{\Sigma}$ with divisor n	$\mathbb{R}^p$, $\mathbb{R}^{p\times p}$
X_c	centred data matrix $X - \mathbf{1}\hat{\mu}^\top$	$\mathbb{R}^{n\times p}$
$\ell(\mu, \Sigma)$	log-likelihood of the training data	scalar
I	identity matrix	$\mathbb{R}^{p\times p}$
$\mathrm{tr}(\cdot)$, $\lvert\cdot\rvert$	trace and determinant of a square matrix	scalar
Eigendecomposition and whitening		
Q	orthogonal matrix whose columns $q_1, \ldots, q_p$ are eigenvectors of Σ	$\mathbb{R}^{p\times p}$
Λ	$\mathrm{diag}(\lambda_1, \ldots, \lambda_p)$, the eigenvalues of Σ	$\mathbb{R}^{p\times p}$
$\Sigma^{1/2}$, $\Sigma^{-1/2}$	symmetric square root and inverse square root of Σ	$\mathbb{R}^{p\times p}$
W	whitening matrix $W = \hat{\Sigma}^{-1/2}$	$\mathbb{R}^{p\times p}$
z	whitened vector $\Sigma^{-1/2}(x - \mu)$	$\mathbb{R}^p$
Z	whitened data matrix	$\mathbb{R}^{n\times p}$ or $\mathbb{R}^{m\times p}$
Detector		
$d^2(x)$	squared Mahalanobis distance using the true μ, Σ	scalar
$\hat{d}^2(x)$	squared Mahalanobis distance using $\hat{\mu}, \hat{\Sigma}$	scalar
χ^2_p, F_p	chi-squared distribution with p degrees of freedom, and its CDF	
$F_{p,\,n-p}$	F distribution with p and $n - p$ degrees of freedom	
α	target false-alarm rate	$(0, 1)$
τ	detection threshold $\tau = F_p^{-1}(1 - \alpha)$	scalar
δ	mean shift of an attack, $x \sim \mathcal{N}(\mu + \delta, \Sigma)$	$\mathbb{R}^p$

3 Definitions and the Link between Data and Math

3.1 Connecting the data to the math

For one user the data matrix is $X \in \mathbb{R}^{n \times p}$. In the benchmark every user has $n = 400$ attempts.

- **One row** $x_i^\top$ is one attempt at typing the password `.tie5Roanl`: the p timings of that attempt.
- **One column** is one timing feature across all attempts, for example the hold time of the first key (`H.period`).
- **One entry** X_{ij} is the value of feature j in attempt i, in seconds.

Attempts of the genuine user are modelled as draws from $\mathcal{N}(\mu, \Sigma)$. Attempts by the other 50 users play the role of anomalies.

3.2 Formal definitions

Definition 1 (Mean vector and covariance matrix)**.** For a random vector $x \in \mathbb{R}^p$, the mean vector is $\mu = \mathbb{E}[x]$ and the covariance matrix is $\Sigma = \mathrm{Cov}(x) = \mathbb{E}[(x-\mu)(x-\mu)^\top]$, so $\Sigma_{jk} = \mathrm{Cov}(x_j, x_k)$.

Definition 2 (Positive semi-definite and positive definite)**.** A symmetric matrix $A \in \mathbb{R}^{p \times p}$ is positive semi-definite (PSD) if $v^\top A v \geq 0$ for all $v \in \mathbb{R}^p$, and positive definite (PD) if $v^\top A v > 0$ for all $v \neq 0$.

Definition 3 (Linear independence, rank and null space)**.** Vectors $a_1, \ldots, a_k$ are linearly independent if $\sum_j c_j a_j = 0$ only when every $c_j = 0$. The rank of a matrix is the number of linearly independent columns. The null space of A is $\{v : Av = 0\}$. A square matrix is invertible if and only if its null space is $\{0\}$.

Definition 4 (Eigenvalues, eigenvectors and the spectral theorem)**.** A scalar λ and a vector $q \neq 0$ with $Aq = \lambda q$ are an eigenvalue and an eigenvector of A. A matrix Q is orthogonal if $Q^\top Q = I$. Spectral theorem (used without proof): a real symmetric Σ can be written $\Sigma = Q \Lambda Q^\top$ with Q orthogonal and Λ diagonal and real. Σ is PD if and only if every $\lambda_i > 0$, and PSD if and only if every $\lambda_i \geq 0$.

Definition 5 (Matrix square roots)**.** For PD $\Sigma = Q \Lambda Q^\top$, $\Sigma^{1/2} = Q \Lambda^{1/2} Q^\top$ and $\Sigma^{-1/2} = Q \Lambda^{-1/2} Q^\top$, where $\Lambda^{\pm 1/2} = \mathrm{diag}(\lambda_i^{\pm 1/2})$.

Definition 6 (Condition number)**.** For PD Σ, the condition number is $\lambda_{\max}/\lambda_{\min}$. A large value means Σ^{-1} is sensitive to small errors in Σ.

Definition 7 (Multivariate normal distribution)**.** For PD Σ, $x \sim \mathcal{N}(\mu, \Sigma)$ has density

$$f(x) = (2\pi)^{-p/2} |\Sigma|^{-1/2} \exp\Big(-\tfrac{1}{2} (x-\mu)^\top \Sigma^{-1} (x-\mu) \Big).$$

Definition 8 (Likelihood and maximum-likelihood estimate)**.** For independent, identically distributed (i.i.d.) $x_1, \ldots, x_n$ with density f, the log-likelihood is $\ell(\mu, \Sigma) = \sum_{i=1}^n \log f(x_i)$. The maximum-likelihood estimate (MLE) is the pair (μ, Σ) that maximises ℓ over $\mu \in \mathbb{R}^p$ and PD Σ.

Definition 9 (Mahalanobis distance)**.** For PD Σ, the squared Mahalanobis distance is $d^2(x) = (x-\mu)^\top \Sigma^{-1} (x-\mu)$. With estimates it is $\hat{d}^2(x) = (x-\hat{\mu})^\top \hat{\Sigma}^{-1} (x-\hat{\mu})$.

Definition 10 (Chi-squared distribution)**.** If $Z_1, \ldots, Z_p$ are independent $\mathcal{N}(0,1)$, then $\sum_i Z_i^2 \sim \chi_p^2$. Its CDF F_p is continuous and strictly increasing on $[0, \infty)$, so the quantile $F_p^{-1}(1-\alpha)$ is unique for $0 < \alpha < 1$. If instead $Z_i \sim \mathcal{N}(c_i, 1)$, the sum has a non-central χ_p^2 distribution with non-centrality $\sum_i c_i^2$.

Definition 11 (Detector, false-alarm rate and detection rate)**.** The detector flags x as an anomaly if $d^2(x) > \tau$. The false-alarm rate is $\mathbb{P}(d^2(x) > \tau)$ when x comes from the genuine user. The detection rate is the same probability when x comes from another user.

3.3 The raw features are linearly dependent

Proposition 1. *For any data matrix $X \in \mathbb{R}^{n \times p}$ and $v \in \mathbb{R}^p$, $v^\top \hat{\Sigma} v = \frac{1}{n}\|X_c v\|^2$. So $\hat{\Sigma}$ is PSD, and $\hat{\Sigma}$ is singular if and only if the columns of X_c are linearly dependent. For the keystroke data with all 31 features,* $\operatorname{rank} \hat{\Sigma} \leq 21$.

Proof. $v^\top \hat{\Sigma} v = \frac{1}{n} v^\top X_c^\top X_c v = \frac{1}{n}\|X_c v\|^2 \geq 0$ [definition of $\hat{\Sigma}$, $(AB)^\top = B^\top A^\top$]. If some $v \neq 0$ has $X_c v = 0$, then $\hat{\Sigma} v = \frac{1}{n} X_c^\top X_c v = 0$, so $\hat{\Sigma}$ has a non-zero null space and is singular. Conversely, if $\hat{\Sigma} v = 0$ for some $v \neq 0$, then $\frac{1}{n}\|X_c v\|^2 = v^\top \hat{\Sigma} v = 0$, so $X_c v = 0$.

For the data, every key pair (a, b) satisfies $\mathrm{DD}(a,b) = \mathrm{H}(a) + \mathrm{UD}(a,b)$ in every row. Let $v^{(ab)}$ have $+1$ on $\mathrm{DD}(a,b)$, -1 on $\mathrm{H}(a)$ and -1 on $\mathrm{UD}(a,b)$. Then $X v^{(ab)} = 0$, and also $X_c v^{(ab)} = 0$ because $X_c v = X v - \mathbf{1}(\hat{\mu}^\top v)$ and $\hat{\mu}^\top v^{(ab)} = 0$. The 10 vectors $v^{(ab)}$ are linearly independent, since each has its $+1$ in a different DD column. So the null space of $\hat{\Sigma}$ has dimension at least 10 and $\operatorname{rank} \hat{\Sigma} \leq 31 - 10 = 21$ [rank-nullity theorem]. □

We keep the 11 H and the 10 UD columns and drop the 10 DD columns, which gives $p = 21$.

4 Assumptions

For each assumption we state what goes wrong in the math if it fails, and how we will check it on the real data in Iteration 3.

A1. The attempts are i.i.d. The attempts $x_1, \ldots, x_n$ of the genuine user are independent draws from the same distribution.
If it fails: the likelihood is no longer a product $\prod_i f(x_i)$, so Theorem 1 does not apply, and the estimates are less precise than n suggests.
Check: plot feature means by session and by repetition, and compute the autocorrelation of each feature over consecutive attempts.

A2. The features are multivariate normal. Genuine attempts follow $\mathcal{N}(\mu, \Sigma)$, possibly after a log transform of the hold times. UD times cannot be log-transformed because some are negative.
If it fails: d^2 no longer follows χ_p^2 (Theorem 3), so the threshold τ no longer gives false-alarm rate α (Theorem 5). In the simulation, heavy-tailed data gave about 5 times too many false alarms, and one real user already gave 0.15 instead of 0.01.
Check: Q-Q plot of d^2 against χ_p^2, a Kolmogorov-Smirnov test, and histograms of the features.

A3. Σ and $\hat{\Sigma}$ are positive definite.
If it fails: Σ^{-1} and $\Sigma^{-1/2}$ do not exist, so d^2 is undefined and the likelihood has no maximum (Theorem 1(c)). This fails for the raw 31 features (Proposition 1).
Check: compute the eigenvalues and the condition number of $\hat{\Sigma}$ for every user. For user `s002` with $p = 21$ the smallest eigenvalue is 7.94×10^{-5} and the condition number is 860.

A4. There are many more attempts than features ($n \gg p$).
If it fails: if $n \leq p$, then $\hat{\Sigma}$ has rank at most $n - 1 < p$ and is singular. For larger but small n, $\hat{\mu}$ and $\hat{\Sigma}$ are noisy, and the χ^2 threshold gives more false alarms than α (Theorem 5(b)). The simulation gives 0.034 at $n = 50$ for a target of 0.01.
Check: here $n = 400$ and $p = 21$. We will also refit with fewer attempts and compare false-alarm rates.

A5. The genuine user types the same way over time. Training and test attempts come from the same distribution.
If it fails: later attempts by the genuine user look like anomalies, so the false-alarm rate rises.
Check: train on early sessions and measure false alarms on each later session separately.

A6. Other users differ from the genuine user in mean or covariance.
If it fails: their d^2 values look like the genuine user's, so the detection rate is low (Theorem 5(c)).
Check: measure the detection rate against each of the other 50 users.

5 Iteration 2 – Math Questions

Theorem k answers Question k. The rule used at each step is given in square brackets.

5.1 Question 1: log-likelihood and the MLEs

Theorem 1. *Let $x_1, \dots, x_n \in \mathbb{R}^p$ be i.i.d. $\mathcal{N}(\mu, \Sigma)$ with PD Σ, and let $\hat{\mu} = \bar{x}$ and $\hat{\Sigma} = \frac{1}{n}\sum_i (x_i - \bar{x})(x_i - \bar{x})^\top$.*

(a) $\ell(\mu, \Sigma) = -\frac{np}{2}\log 2\pi - \frac{n}{2}\log|\Sigma| - \frac{1}{2}\sum_i (x_i - \mu)^\top \Sigma^{-1}(x_i - \mu)$.
(b) If $\hat{\Sigma}$ is PD, then $(\hat{\mu}, \hat{\Sigma})$ is the unique maximiser of ℓ.
(c) If $\hat{\Sigma}$ is singular, ℓ is unbounded above, so no MLE exists.

Proof. (a) By independence, $\ell = \sum_i \log f(x_i)$ [definition of likelihood]. Taking the log of the normal density and adding the n terms gives (a) [$\log(ab) = \log a + \log b$].

Step 1: split the quadratic form. Write $x_i - \mu = (x_i - \bar{x}) + (\bar{x} - \mu)$ and expand. The cross terms contain $\sum_i (x_i - \bar{x}) = 0$ [definition of $\bar{x}$], so

$$\sum_i (x_i - \mu)(x_i - \mu)^\top = n\hat{\Sigma} + n(\bar{x} - \mu)(\bar{x} - \mu)^\top.$$

Since $a^\top M a = \operatorname{tr}(Maa^\top)$ [a scalar equals its trace; cyclic property of the trace],

$$\sum_i (x_i - \mu)^\top \Sigma^{-1}(x_i - \mu) = n\,\operatorname{tr}(\Sigma^{-1}\hat{\Sigma}) + n(\bar{x} - \mu)^\top \Sigma^{-1}(\bar{x} - \mu).$$

Hence

$$\ell(\mu, \Sigma) = -\tfrac{np}{2}\log 2\pi - \tfrac{n}{2}\big[\log|\Sigma| + \operatorname{tr}(\Sigma^{-1}\hat{\Sigma})\big] - \tfrac{n}{2}(\bar{x} - \mu)^\top \Sigma^{-1}(\bar{x} - \mu). \tag{1}$$

Step 2: the mean. Σ^{-1} is PD (Theorem 2), so the last term of (1) is ≤ 0, with equality if and only if $\mu = \bar{x}$ [definition of PD]. This holds for every Σ, so $\hat{\mu} = \bar{x}$.

Step 3: the covariance, part (b). With $\mu = \bar{x}$ we must minimise $g(\Sigma) = \log|\Sigma| + \operatorname{tr}(\Sigma^{-1}\hat{\Sigma})$ over PD Σ. Since $\hat{\Sigma}$ is PD, $\hat{\Sigma}^{1/2}$ and $\hat{\Sigma}^{-1/2}$ exist [spectral theorem]. Let $A = \hat{\Sigma}^{-1/2}\Sigma\hat{\Sigma}^{-1/2}$. It is PD with eigenvalues $a_1, \dots, a_p > 0$ [$v^\top A v = (\hat{\Sigma}^{-1/2}v)^\top \Sigma (\hat{\Sigma}^{-1/2}v) > 0$], and $\Sigma = \hat{\Sigma}^{1/2} A \hat{\Sigma}^{1/2}$. Then

$$\log|\Sigma| = \log|\hat{\Sigma}| + \log|A| = \log|\hat{\Sigma}| + \textstyle\sum_i \log a_i \qquad [\det(BC) = \det B \det C;\ \det = \textstyle\prod \text{eigenvalues}]$$
$$\operatorname{tr}(\Sigma^{-1}\hat{\Sigma}) = \operatorname{tr}(\hat{\Sigma}^{-1/2}A^{-1}\hat{\Sigma}^{-1/2}\hat{\Sigma}) = \operatorname{tr}(A^{-1}) = \textstyle\sum_i 1/a_i \quad [\text{cyclic property of the trace}].$$

So $g(\Sigma) = \log|\hat{\Sigma}| + \sum_i h(a_i)$ with $h(a) = \log a + 1/a$. Since $h'(a) = (a-1)/a^2$ is negative for $a < 1$ and positive for $a > 1$, h has its unique minimum $h(1) = 1$ on $(0, \infty)$ [first-derivative test]. So $g(\Sigma) \ge \log|\hat{\Sigma}| + p$, with equality if and only if every $a_i = 1$, that is $A = I$, that is $\Sigma = \hat{\Sigma}$.

(c) Let $\hat{\Sigma}$ have eigenvalues $\lambda_i \ge 0$ with at least one $\lambda_i = 0$. Take $\mu = \bar{x}$ and $\Sigma_\varepsilon = \hat{\Sigma} + \varepsilon I$, which is PD for $\varepsilon > 0$. By (1), $\ell = \text{const} - \frac{n}{2}\big[\sum_i \log(\lambda_i + \varepsilon) + \sum_i \frac{\lambda_i}{\lambda_i + \varepsilon}\big]$. As $\varepsilon \to 0$, the term with $\lambda_i = 0$ is $\log\varepsilon \to -\infty$, the other logs stay bounded and the second sum is at most p. So $\ell \to +\infty$. □

5.2 Question 2: the Mahalanobis distance is non-negative

Theorem 2. *If Σ is PD, then Σ^{-1} is PD, $d^2(x) \geq 0$ for all x, and $d^2(x) = 0$ if and only if $x = \mu$.*

Proof. $\Sigma = Q\Lambda Q^\top$ with every $\lambda_i > 0$ [spectral theorem]. Then $\Sigma^{-1} = Q\Lambda^{-1}Q^\top$ [check: $(Q\Lambda Q^\top)(Q\Lambda^{-1}Q^\top) = I$ using $Q^\top Q = I$], whose eigenvalues $1/\lambda_i$ are positive, so Σ^{-1} is PD. Let $y = Q^\top(x - \mu)$. Then

$$d^2(x) = (x-\mu)^\top Q\Lambda^{-1}Q^\top(x-\mu) = y^\top\Lambda^{-1}y = \sum_{i=1}^{p}\frac{y_i^2}{\lambda_i} \geq 0,$$

a sum of non-negative terms. It is 0 if and only if $y = 0$, that is $x = \mu$ [Q is invertible]. □

5.3 Question 3: d^2 follows a χ^2_p distribution

Theorem 3. *Let $x \sim \mathcal{N}(\mu, \Sigma)$ with PD Σ and $z = \Sigma^{-1/2}(x - \mu)$. Then $z_1, \ldots, z_p$ are i.i.d. $\mathcal{N}(0,1)$ and $d^2(x) = \|z\|^2 \sim \chi^2_p$.*

Proof. Step 1: $d^2 = \|z\|^2$. $\Sigma^{-1/2}$ is symmetric and $(\Sigma^{-1/2})^2 = \Sigma^{-1}$ (Theorem 4), so $\|z\|^2 = (x-\mu)^\top\Sigma^{-1/2}\Sigma^{-1/2}(x-\mu) = d^2(x)$.

Step 2: the density of z. The map $x = \mu + \Sigma^{1/2}z$ is linear and invertible with Jacobian determinant $|\Sigma|^{1/2}$ [$\det\Sigma^{1/2} = \prod_i\sqrt{\lambda_i}$]. By the change-of-variables formula,

$$f_z(z) = f_x(\mu + \Sigma^{1/2}z)\,|\Sigma|^{1/2} = (2\pi)^{-p/2}\exp\Big(-\tfrac{1}{2}z^\top\Sigma^{1/2}\Sigma^{-1}\Sigma^{1/2}z\Big) = \prod_{i=1}^{p}\frac{1}{\sqrt{2\pi}}e^{-z_i^2/2},$$

using $\Sigma^{1/2}\Sigma^{-1}\Sigma^{1/2} = I$. The joint density is a product of p standard normal densities, so the z_i are independent $\mathcal{N}(0,1)$ [a joint density that factors implies independence].

Step 3. By the definition of χ^2_p, $\|z\|^2 = \sum_i z_i^2 \sim \chi^2_p$. □

Remark. It follows that $\mathbb{E}[d^2] = p$ and $\mathrm{Var}(d^2) = 2p$. The simulation checks both.

5.4 Question 4: $\Sigma^{-1/2}$ and its geometric meaning

Theorem 4. *Let $\Sigma = Q\Lambda Q^\top$ be PD and $W = \Sigma^{-1/2} = Q\Lambda^{-1/2}Q^\top$. Then:*

- *(a) W is symmetric and PD, $W^2 = \Sigma^{-1}$, and $W\Sigma W = I$.*
- *(b) $d^2(x) = \|W(x-\mu)\|^2$, and if $x \sim \mathcal{N}(\mu, \Sigma)$ then $\mathrm{Cov}(W(x-\mu)) = I$.*
- *(c) The set $\{x : d^2(x) \leq c\}$ is an ellipsoid centred at μ, with axes along $q_1, \ldots, q_p$ and semi-axis lengths $\sqrt{c\lambda_i}$. W maps it onto the ball of radius $\sqrt{c}$.*

Proof. (a) $W^\top = Q\Lambda^{-1/2}Q^\top = W$ [$(ABC)^\top = C^\top B^\top A^\top$; Λ is diagonal]. Its eigenvalues $\lambda_i^{-1/2}$ are positive, so W is PD. Using $Q^\top Q = I$,

$$W^2 = Q\Lambda^{-1/2}(Q^\top Q)\Lambda^{-1/2}Q^\top = Q\Lambda^{-1}Q^\top = \Sigma^{-1},$$
$$W\Sigma W = Q\Lambda^{-1/2}Q^\top\, Q\Lambda Q^\top\, Q\Lambda^{-1/2}Q^\top = Q\Lambda^{-1/2}\Lambda\Lambda^{-1/2}Q^\top = I.$$

(b) $\|W(x-\mu)\|^2 = (x-\mu)^\top W^\top W(x-\mu) = (x-\mu)^\top \Sigma^{-1}(x-\mu)$. Also $\mathrm{Cov}(Wu) = W\,\mathrm{Cov}(u)W^\top$ [linearity of expectation], so $\mathrm{Cov}(W(x-\mu)) = W\Sigma W = I$.

(c) Let $y = Q^\top(x-\mu)$, the coordinates of $x-\mu$ along the eigenvectors. Then $d^2(x) \le c$ if and only if $\sum_i \left(y_i/\sqrt{c\lambda_i}\right)^2 \le 1$, which is an ellipsoid with semi-axes $\sqrt{c\lambda_i}$ in the directions q_i.

Geometric meaning. $W(x-\mu) = Q\Lambda^{-1/2}Q^\top(x-\mu)$ does three things: $Q^\top$ rotates the ellipsoid so its axes line up with the coordinate axes, $\Lambda^{-1/2}$ shrinks or stretches axis i by $1/\sqrt{\lambda_i}$ so every axis has the same length, and Q rotates back. Rotations keep lengths, so the ellipsoid becomes the ball $\{u : \|u\|^2 \le c\}$. This is why W is called a whitening matrix: it turns correlated data with unequal spreads into uncorrelated data with unit spread. □

Corollary 1. *If $\hat\Sigma$ is PD and $W = \hat\Sigma^{-1/2}$, the whitened training data $Z = X_cW$ satisfy $\frac{1}{n}Z^\top Z = I$ exactly, and the mean of $\hat d^2$ over the training data equals p.*

Proof. $\frac{1}{n}Z^\top Z = \frac{1}{n}W^\top X_c^\top X_c W = W\hat\Sigma W = I$ by (a) applied to $\hat\Sigma$. The mean of the squared row norms is $\frac{1}{n}\,\mathrm{tr}(ZZ^\top) = \mathrm{tr}(\frac{1}{n}Z^\top Z) = \mathrm{tr}(I) = p$. □

5.5 Question 5: the χ^2 threshold gives false-alarm rate α

Theorem 5. *Let $x \sim \mathcal{N}(\mu, \Sigma)$ with μ, Σ known and Σ PD, and set $\tau = F_p^{-1}(1-\alpha)$. Then:*

- *(a)* $\mathbb{P}(d^2(x) > \tau) = \alpha$.
- *(b) (Estimated parameters, stated without proof.) If $\hat\mu, \hat\Sigma$ come from n independent genuine attempts and x is a new one, then $\hat d^2(x) \sim \frac{p(n+1)}{n-p}F_{p,\,n-p}$ (Hotelling's T^2). The false-alarm rate is then above α and tends to α as $n \to \infty$.*
- *(c) (Mean shift.) If $x \sim \mathcal{N}(\mu+\delta, \Sigma)$, then $d^2(x)$ is non-central χ^2_p with non-centrality $\delta^\top\Sigma^{-1}\delta$, and the detection rate is the probability that it exceeds τ.*

Proof. (a) By Theorem 3, $d^2(x)$ has CDF F_p. So $\mathbb{P}(d^2 > \tau) = 1 - F_p(\tau) = 1 - F_p\big(F_p^{-1}(1-\alpha)\big) = 1-(1-\alpha) = \alpha$ [F_p is continuous and strictly increasing, so $F_p(F_p^{-1}(u)) = u$].

(c) Now $z = \Sigma^{-1/2}(x-\mu) \sim \mathcal{N}(\Sigma^{-1/2}\delta, I)$ [Theorem 4(b) and the shift by δ], so $\|z\|^2$ is non-central χ^2_p with non-centrality $\|\Sigma^{-1/2}\delta\|^2 = \delta^\top\Sigma^{-1}\delta$ [definition]. □

6 The Algorithm as Math

We train on $X \in \mathbb{R}^{n \times p}$ (one genuine user) and then score the new attempts $Y \in \mathbb{R}^{m \times p}$. Operation counts are multiply-adds to leading order. The last column uses the real data: $n = 400$, $p = 21$, and one new attempt.

Step	Equation	Matrix sizes	Operations	n=400, p=21
1	$\hat{\mu} = \frac{1}{n} X^\top \mathbf{1}$	$(p \times n)(n \times 1) \to p \times 1$	np	8,400
2	$X_c = X - \mathbf{1}\hat{\mu}^\top$	$n \times p$	np	8,400
3	$\hat{\Sigma} = \frac{1}{n} X_c^\top X_c$	$(p \times n)(n \times p) \to p \times p$	np^2	176,400
4	$\hat{\Sigma} = Q \Lambda Q^\top$	Q, Λ: $p \times p$	$O(p^3)$	about 10^4
5	check $\lambda_{\min} > 0$ (A3), else stop	p eigenvalues	p	21
6	$W = Q \Lambda^{-1/2} Q^\top$	$(p \times p)(p \times p)$	p^3	9,261
7	$\tau = F_p^{-1}(1 - \alpha)$	scalar	one quantile	
8	$Z = (Y - \mathbf{1}\hat{\mu}^\top) W$	$(m \times p)(p \times p) \to m \times p$	mp^2	441
9	$\hat{d}_j^2 = \sum_k Z_{jk}^2$	$m \times 1$	mp	21
10	flag attempt j if $\hat{d}_j^2 > \tau$	m comparisons	m	1

Training costs $O(np^2 + p^3)$ and is dominated by step 3. Scoring costs $O(p^2)$ per new attempt. Steps 1 to 3 are the formulas of Theorem 1, steps 4 to 6 are Theorem 4, and steps 7 to 10 use Theorems 2, 3 and 5.

7 Real Data

Size. 20,400 rows and 34 columns: 51 subjects, each typing the password `.tie5Roanl` 400 times (8 sessions of 50 repetitions). Every subject has exactly 400 rows. **Missing values:** none.

Variables	Count	Type	Meaning
`subject`	1	string	subject ID (`s002`, ...)
`sessionIndex`, `rep`	2	integer	session (1 to 8), repetition in the session (1 to 50)
`H.*`	11	float (s)	hold time: key down to the same key up
`DD.*`	10	float (s)	down-down: key down to next key down
`UD.*`	10	float (s)	up-down: key up to next key down

Feature (all subjects)	Mean	Std	Min	Max	Skewness
`H.period`	0.0934	0.0296	0.0014	0.3761	1.27
`H.Return`	0.0883	0.0275	0.0029	0.2651	0.73
`DD.period.t`	0.2641	0.2205	0.0187	12.5061	12.11
`UD.period.t`	0.1708	0.2268	−0.2358	12.4517	11.35

Observations.

1. **Hold times are right-skewed.** Their median skewness is 0.90, and -0.50 after a log transform (Figure 1). We will compare raw and log hold times in Iteration 3.
2. **Some up-down times are negative.** 22,118 UD entries (about 10.8%) are negative, because the next key is sometimes pressed before the previous key is released. So the log is defined for H and DD but not for UD.
3. **The features are linearly dependent.** DD = H + UD holds in every row (largest error 8.9×10^{-16}). For subject `s002` with all 31 features, the covariance has rank 21 and 10 eigenvalues are numerically zero (Figure 3). With the 21 features H and UD, the smallest eigenvalue is 7.94×10^{-5} and the condition number is 860.

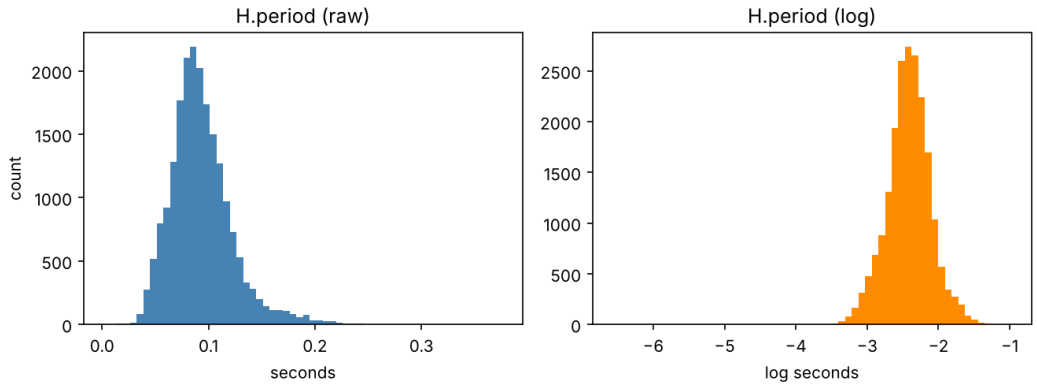


Figure 1: Hold time `H.period` for all subjects, raw (left) and log (right).

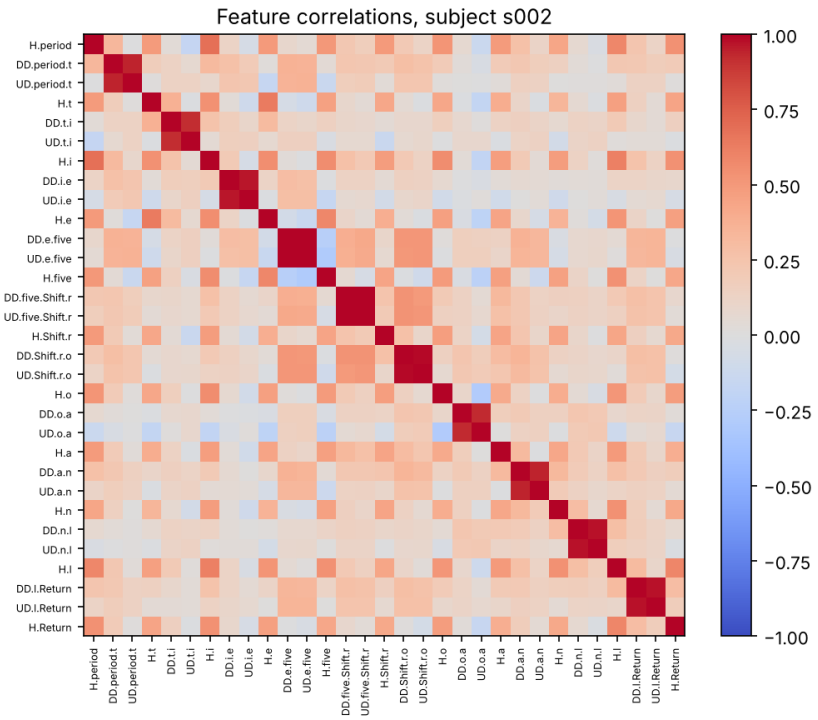


Figure 2: Correlations between the 31 features for subject `s002`.

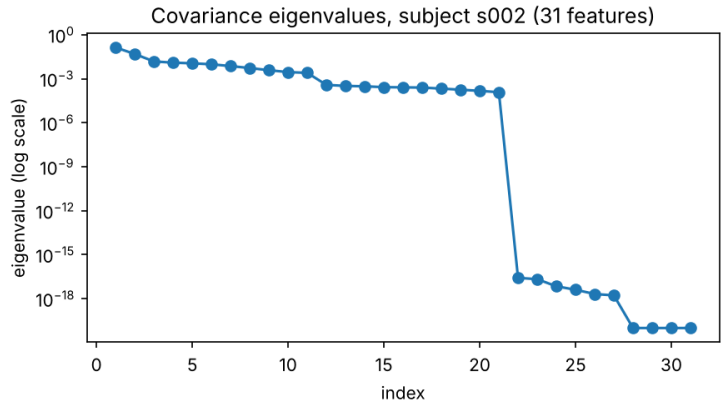


Figure 3: Eigenvalues of the sample covariance of `s002` with all 31 features. The last ten are numerically zero.

Next step (Iteration 3). Use the 21 features H and UD, decide on the hold-time transform, fit $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$ on each user's early sessions, measure false alarms on that user's later sessions and detections on the other users, and check assumptions A1 to A6.

8 Simulation Check

8.1 Model and parameters

All numbers come from `src/simulation.py` with seed `np.random.default_rng(0)`. We draw $n = 5000$ points in $p = 5$ dimensions from $\mathcal{N}(\mu, \Sigma)$ with $\mu = 0$ and $\Sigma = BB^\top + 5I$, where B is a 5×5 matrix of standard normal entries. We use $\alpha = 0.01$, so $\tau = \chi^2_{5,0.99} = 15.0863$, and add 50 attacks from $\mathcal{N}(\mu + 4 \cdot \mathbf{1}, \Sigma)$.

$$\Sigma = \begin{pmatrix} 5.74 & 1.08 & -0.93 & -0.74 & -1.07 \\ 1.08 & 9.82 & -0.64 & -2.88 & -0.29 \\ -0.93 & -0.64 & 12.40 & -0.22 & 0.48 \\ -0.74 & -2.88 & -0.22 & 7.19 & 0.65 \\ -1.07 & -0.29 & 0.48 & 0.65 & 8.27 \end{pmatrix}, \qquad \hat{\Sigma} = \begin{pmatrix} 5.78 & 1.20 & -0.91 & -0.92 & -1.10 \\ 1.20 & 9.84 & -0.60 & -2.91 & -0.25 \\ -0.91 & -0.60 & 12.21 & -0.02 & 0.67 \\ -0.92 & -2.91 & -0.02 & 7.15 & 0.50 \\ -1.10 & -0.25 & 0.67 & 0.50 & 8.13 \end{pmatrix}.$$

The eigenvalues of Σ are 5.11, 5.24, 8.37, 11.62 and 13.08, all positive, so Σ is PD.

8.2 True values and estimates

Quantity	True	Estimate
$\mu_1,\ \mu_2,\ \mu_3,\ \mu_4,\ \mu_5$	0, 0, 0, 0, 0	0.053, 0.001, 0.023, 0.009, −0.015
largest entry error $\max_{jk} \lvert\hat{\Sigma}_{jk} - \Sigma_{jk}\rvert$		0.2034
relative error $\lVert\hat{\Sigma} - \Sigma\rVert_F / \lVert\Sigma\rVert_F$		0.0290
mean of d^2	5	5.000
variance of d^2	10	10.20
false-alarm rate	0.0100	0.0102
detection rate on the 50 attacks	0.60	0.54

The true detection rate 0.60 is the probability that a non-central χ^2_5 with non-centrality $\delta^\top \Sigma^{-1} \delta = 13.0$ exceeds τ (Theorem 5(c)). With only 50 attacks its standard error is about 0.07, so 0.54 agrees.

8.3 What the simulation confirms

- **Theorem 1:** $\hat{\mu}$ and $\hat{\Sigma}$ are close to μ and Σ (relative error 0.029), and $\hat{\Sigma}$ equals `np.cov(X, bias=True)`.
- **Theorem 2:** all 5000 values of d^2 are non-negative (the smallest is 0.060).
- **Theorem 3:** d^2 has variance 10.20 against $2p = 10$, its histogram follows the χ^2_5 density (Figure 4), and a Kolmogorov-Smirnov test does not reject χ^2_5 (p-value 0.47). The mean of exactly 5 is guaranteed by Corollary 1.
- **Theorem 4:** the whitened data have covariance I up to 2×10^{-15}, and W is symmetric with $W^2 \hat{\Sigma} = I$.
- **Theorem 5:** the false-alarm rate is 0.0102 against $\alpha = 0.01$, within one standard error (0.0014).

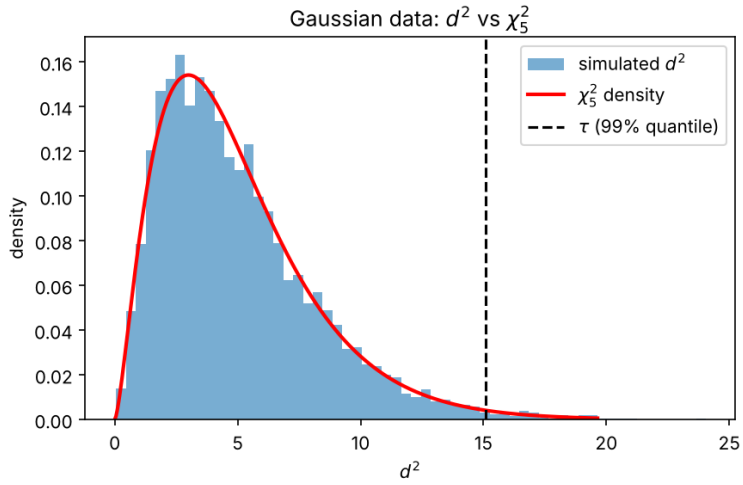


Figure 4: Normal data: histogram of d^2 with the χ^2_5 density and the threshold τ.

8.4 When the assumptions fail

Heavy tails (A2). We redraw the data from a multivariate t distribution with 3 degrees of freedom, scaled to the same mean and covariance. Whitening still works, because it only uses the mean and covariance, but d^2 no longer follows χ^2_5. The false-alarm rate is 0.057 against 0.01, and the Kolmogorov-Smirnov p-value is numerically 0. Over 200 repetitions the false-alarm rate is 0.0100 ± 0.0012 for normal data and 0.0526 ± 0.0042 for t data, about 5 times too high (Figure 5).

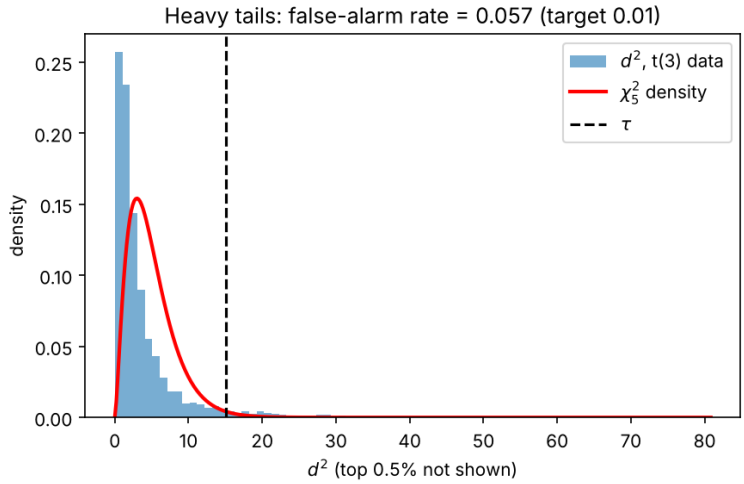


Figure 5: Heavy-tailed t data: histogram of d^2 with the χ^2_5 density.

Small samples (A4). We fit $\hat{\mu}, \hat{\Sigma}$ on n points and measure the false-alarm rate on new points (200 repetitions for each n, target 0.01). The last column is the exact value from Theorem 5(b).

n	relative error of $\hat{\Sigma}$	simulated false-alarm rate	Theorem 5(b)
50	0.3156	0.0342	0.0343
100	0.2286	0.0183	0.0197
400	0.1143	0.0123	0.0120
1000	0.0712	0.0105	0.0108
5000	0.0323	0.0102	0.0102

The error of $\hat{\Sigma}$ falls roughly like $1/\sqrt{n}$, and Theorem 5(b) explains the extra false alarms at small n (Figure 6).

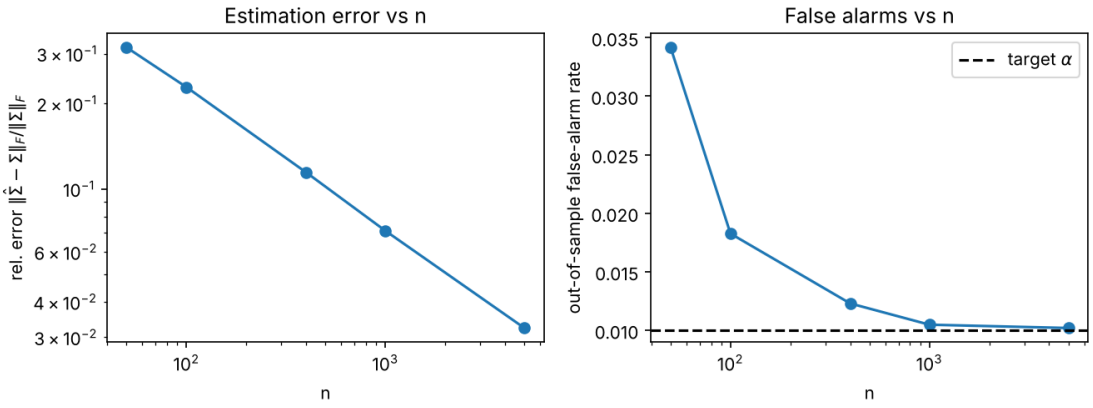


Figure 6: Left: relative error of $\hat{\Sigma}$ against n (log scales). Right: false-alarm rate on new points against n, approaching the target $\alpha = 0.01$.

Singular covariance (A3). Adding a column equal to the sum of two others gives a smallest eigenvalue of 1.5×10^{-15}, and our function `inv_sqrt_psd` stops with an error. The real data behave the same way: with all 31 features the estimate for `s002` is not PD, and with the 21 features H and UD it is, with mean $\hat{d}^2 = 21.00 = p$ as Corollary 1 predicts.

First look at real data (A2). With these 21 features, 0.150 of `s002`'s own attempts have $\hat{d}^2 > \tau$, against a target of 0.01. This is one user, measured on the training data without a log transform, but it suggests the real timings are not normal. We will test this properly in Iteration 3.

Reference. K. S. Killourhy and R. A. Maxion, *Comparing anomaly-detection algorithms for keystroke dynamics*, IEEE/IFIP International Conference on Dependable Systems and Networks (DSN), 2009.